

SMART PARKING PLATFORM

Homework 4: Risk Management and Communication Plan

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6. Risk Management

6.1 Risk Identification

At least four risks were identified from each of the categories: technical, schedule, financial, and people.

Technical

- Third-party payment gateway API changes or downtime disrupts checkout.
- Real-time occupancy data lags between sensor/LPR feeds and the app.
- Mapping/navigation API hits rate limits or is deprecated.
- Security vulnerability in stored payment or user data.

Schedule

- Backend delays cascade to every dependent dev team.
- Third-party integration onboarding such as payment and mapping takes longer than planned.
- A team member is unavailable during a critical sprint, for example: exams or illness.
- Scope creep from stakeholder requests extends the timeline.

Financial

- Third-party API usage costs exceed the estimated allocation.
- Additional tooling and licensing costs weren't budgeted.
- An external security audit is required, adding an unplanned cost.
- A third-party vendor changes pricing in the middle of the project.

People

- A key contributor like a sole Backend/API developer becomes unavailable.
- Workload is distributed unevenly across teams, risking burnout.
- Miscommunication between teams from unclear ownership.
- Stakeholders are unavailable for timely feedback or approvals.

6.2 Risk Register

All sixteen identified risks are tracked below, exceeding the ten-entry minimum.

ID	Description	Prob.	Impact	Owner	Response Strategy	Status
R-T1	Third-party payment gateway API changes or downtime disrupts checkout	Medium	High	Payment Team	Mitigate: abstract payment calls behind an internal service layer; monitor vendor status page	Open
R-T2	Real-time occupancy data lags between sensor/LPR feeds and the app	Medium	Medium	Backend/API	Mitigate: add caching and a periodic reconciliation job	Open
R-T3	Mapping/navigation API hits rate limits or is deprecated	Low	Medium	Mapping Team	Mitigate: cache map data; evaluate a backup provider	Open

ID	Description	Prob.	Impact	Owner	Response Strategy	Status
R-T4	Security vulnerability in stored payment or user data	Low	High	Backend/API	Avoid: follow security guidance; third-party security review before release	Open
R-S1	Backend/API delays cascade to every dependent dev team	Medium	High	Backend/API	Mitigate: prioritize the API contract in Sprint 1, daily standups	Open
R-S2	Third-party integration onboarding takes longer than planned	Medium	Medium	PM Office	Mitigate: start vendor onboarding early, keep sandbox credentials as a fallback	Open
R-S3	Team member unavailable during a critical sprint	High	Medium	PM Office	Accept: cross-train members, keep shared documentation up-to-date	Open
R-S4	Scope creep from stakeholder requests extends the timeline	Medium	High	PM Office	Mitigate: enforce a change-request process, defer additions to backlog	Open
R-F1	Third-party API usage costs exceed the estimated allocation	Low	Medium	PM Office	Mitigate: monitor usage dashboards, set some billing alerts	Open
R-F2	Additional tooling or licensing costs were not budgeted	Low	Low	PM Office	Accept: review tool needs each sprint, prefer open-source where viable	Open
R-F3	An external security audit is required, adding unplanned cost	Medium	Medium	PM Office	Transfer: hold a contingency allowance for compliance review	Open
R-F4	A third-party vendor changes pricing mid-project	Low	Medium	PM Office	Mitigate: negotiate fixed-rate terms where possible	Open
R-P1	A key contributor becomes unavailable	Medium	High	PM Office	Mitigate: document as you go, cross-train a backup	Open
R-P2	Workload is distributed unevenly across teams, risking burnout	Medium	Medium	PM Office	Mitigate: rebalance tasks during sprint planning	Open
R-P3	Miscommunication between teams from unclear ownership	Medium	Medium	PM Office	Mitigate: clarify ownership per deliverable, keep one shared backlog	Open
R-P4	Stakeholders are unavailable for timely feedback or approvals	Medium	High	PM Office	Mitigate: schedule recurring stakeholder review slots in advance	Open

Table 1: Risk register — ID, description, probability, impact, owner, response strategy, and status.

7. Communication Plan

The table below sets the meeting cadence and reporting method for each team and for stakeholders.

Team / Group	Meeting Cadence	Reporting Method
All Dev Teams (standup)	Daily, 15 minutes	Verbal standup + shared Trello board
Backend/API Team	Weekly sync	Sprint board update + API changelog

Team / Group	Meeting Cadence	Reporting Method
Mobile App Team	Weekly sync	Sprint board update + build status in Trello
Web App Team	Weekly sync	Sprint board update + shared screenshots doc
Mapping & Location Services	Bi-weekly sync	Integration status report
Payment Integration Team	Bi-weekly sync	Integration status + compliance checklist
QA & Testing Team	Weekly sync	Defect log + test coverage report
Team Leads / PM Office	Weekly project status meeting	Written status report + risk register review
Stakeholders (external)	Weekly	Stakeholder update video + written summary in repo

Table 2: Communication plan — meeting cadence and reporting method by team.